Hydrostatic-Based Proofs in Geometry

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Abstract

Inspired by Tokieda's work on mechanical insights in geometry, we explore a hydrostatic approach to classical geometric problems. Using principles of fluid statics, we derive two mathematical results regarding polygons. These results illustrate how physical principles can assist in understanding mathematical identities.

I. INTRODUCTION

The work of Tokieda, Levi, Vallejo & Bove, and Kim shows how physical laws can lead to alternative proofs of mathematical theorems.[1–4] In a similar spirit, in this paper we investigate how hydrostatic equilibrium can offer a novel approach to geometric problems.

We consider a system where rigid plates joined along their edges enclose a fluid. By analyzing the equilibrium of the fluid, we show that the polygon maximizing the enclosed area for fixed side lengths must have its vertices on a common circle; this is done in Section II. In Section III we derive the Shoelace formula for the area of a polygon by considering the forces acting on a submerged polygonal region. These results point out new connections between mathematics and physics.

II. ISOPERIMETRIC POLYGON THEOREM

We use the term 'water' to refer to an incompressible fluid of uniform density ρ. For simplicity, we set the atmospheric pressure to zero, although this assumption is not essential.

Among all simple closed polygons with given side lengths $L_0, L_1, \cdots, L_{n-1}$ in counterclockwise order, the one enclosing the maximum area is cyclic; that is, all its vertices lie on a common circle. There are several mathematical proofs of this theorem, including approaches based on the isoperimetric inequality[5] (Section 12.5), Bretschneider's formula[6] (Lemma 2.4), and calculus methods[7,8]. Here, we present a proof using hydrostatic principles.

See Figure 1. Consider n rigid rectangular plates of widths $L_0, L_1, \cdots, L_{n-1}$. Join these plates along their edges to form a closed polygon, and then partially fill the enclosed region with water. Assume that the plates can move freely. Since the volume of water is constant, its potential energy will be lowest when its area is highest, so the polygon's area will be maximized. (The existence of such a configuration is guaranteed by the compactness of the domain and the continuity of the area.) Denote the position vectors of the polygon's vertices by $\mathbf{r}_0, \mathbf{r}_1, \cdots, \mathbf{r}_{n-1}$ in counterclockwise order, chosen so that the vector

$$\mathbf{L}_i = \mathbf{r}_{i+1} - \mathbf{r}_i \tag{1}$$

has length L_i for each i, as shown in Fig. 1. (We adopt the conventions $\mathbf{r}_i = \mathbf{r}_{i \bmod n}$.) We will refer to the plate corresponding to $\mathbf{L}_i$ as plate P_i.

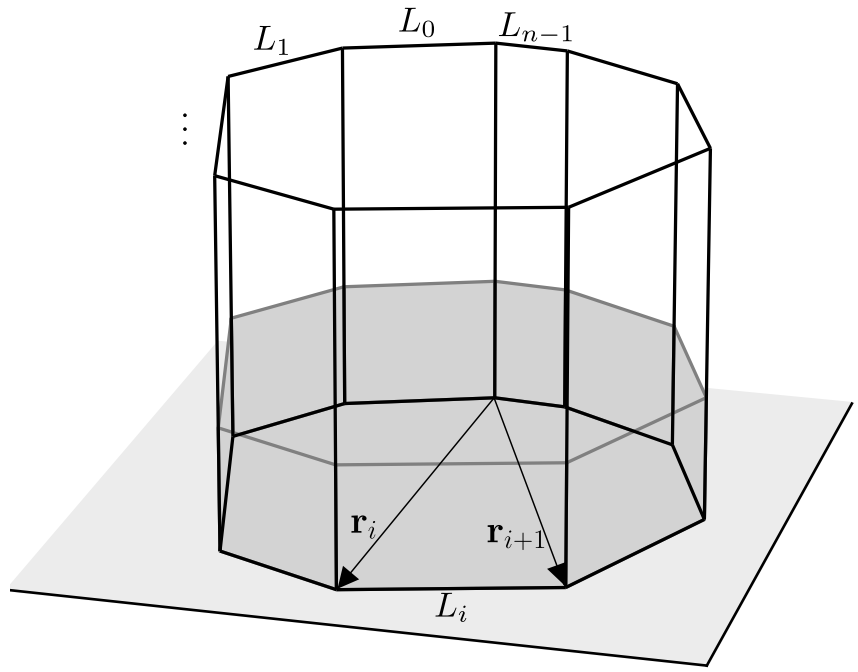

FIG. 1. Picture of the fluid system

Each plate exerts a force on its two neighboring plates. Figure 2 introduces the notation: $\mathbf{N}_i$ is the horizontal component(parallel to the ground) of the force exerted by plate P_i on its adjacent plate P_{i-1}, for all i. By Newton's third law, plate P_{i-1} exerts an equal and opposite force on plate P_i; hence, the horizontal component of this reaction force is $-\mathbf{N}_i$, for all i.

At equilibrium, if the water is of depth h, the average pressure on plate P_i is $p_{i,\mathrm{av}} = \rho g h/2$. Let $\mathbf{F}_i$ denote the force vector exerted by the water on plate P_i. The magnitude of this force is

$$\|\mathbf{F}_i\| = p_{i,\mathrm{av}} A_i = \frac{\rho g h^2}{2} L_i = f L_i, \tag{2}$$

where we have defined $f = \rho g h^2/2$.

Since the system is in equilibrium, both the net force and the net torque on plate P_i must vanish. In particular, the net force along the direction of $\mathbf{L}_i$ is zero, which implies that the components of the forces $\mathbf{N}_i$ and $\mathbf{N}_{i+1}$ in the direction of $\mathbf{L}_i$ are equal. Similarly, by taking the torque about the center of plate P_i, we deduce that the components of $-\mathbf{N}_i$ and $\mathbf{N}_{i+1}$ perpendicular to $\mathbf{L}_i$ must also be equal. Consequently, we obtain

$$\|\mathbf{N}_i\| = \|\mathbf{N}_{i+1}\| \tag{3}$$

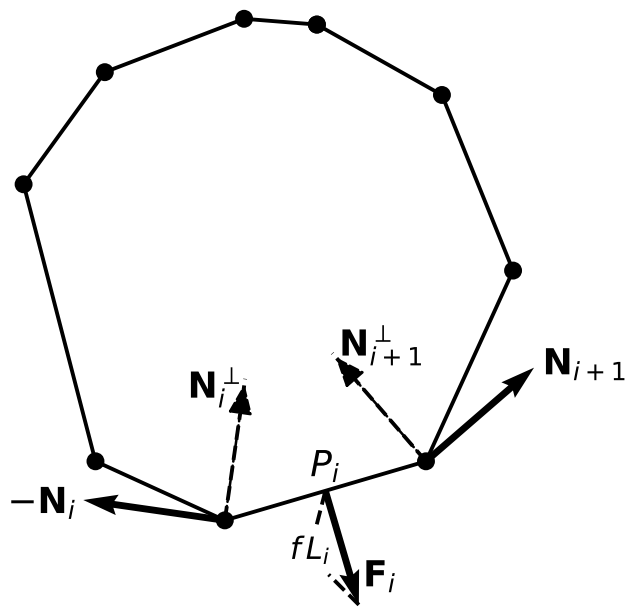

FIG. 2. Force diagram of plate P_i

for all i.

Define $\mathbf{N}_i^{\perp}$ as the vector obtained by rotating $\mathbf{N}_i$ by 90° counterclockwise, for all i. When the force due to the water is rotated by 90°, it becomes $f\mathbf{L}_i$. Since the net force exerted on plate P_i is zero, its rotated counterpart must also sum to zero. Consequently,

$$-\mathbf{N}_i^{\perp} + f\,\mathbf{L}_i + \mathbf{N}_{i+1}^{\perp} = 0. \tag{4}$$

Dividing by f and adding $\mathbf{r}_i + \mathbf{N}_i^{\perp}/f$ to both sides, and then using Eq. (1), this gives

$$\mathbf{r}_i + \frac{1}{f}\mathbf{N}_i^{\perp} = \mathbf{r}_i + \mathbf{L}_i + \frac{1}{f}\mathbf{N}_{i+1}^{\perp} \tag{5}$$

$$= \mathbf{r}_{i+1} + \frac{1}{f}\mathbf{N}_{i+1}^{\perp} \tag{6}$$

for all i.

The equality in Eq. (6) shows that $\mathbf{r}_i + \mathbf{N}_i^{\perp}/f$ is independent of i, and we denote this common vector as $\mathbf{r}_c$. It follows that

$$\|\mathbf{r}_i - \mathbf{r}_c\| = \frac{1}{f}\|\mathbf{N}_i^{\perp}\| = \frac{1}{f}\|\mathbf{N}_i\|, \tag{7}$$

and by Eq. (3), this length is identical for all i. Therefore, every vertex of the polygon lies on the circle centered at $\mathbf{r}_c$. This completes the proof.

III. SHOELACE FORMULA

The Shoelace formula provides an easy method for computing the area of a simple polygon that is non-self-intersecting and encloses a single region without gaps given the Cartesian positions of its vertices. Let $\mathcal{P}$ be an n-sided simple polygon that lies in the xy plane with vertices represented by position vectors $\mathbf{r}_0, \mathbf{r}_1, \cdots, \mathbf{r}_{n-1}$, arranged in counterclockwise order. If $\mathbf{r}_i = (x_i, y_i)$, then the area A of the polygon is given by the Shoelace formula:

$$A = \frac{1}{2}\sum_{i=0}^{n-1}(x_i y_{i+1} - x_{i+1} y_i). \tag{8}$$

We now develop a hydrostatic proof of this formula. Consider a solid prism that has the polygon as its base and has thickness z_0, as shown in Fig. 3. It is submerged in water, with the top surface of the water in the plane $y = 0$. (The submersion may require translating the prism in the negative y direction.) The gravitational acceleration is given by $\mathbf{g} = -g\hat{\mathbf{y}}$.

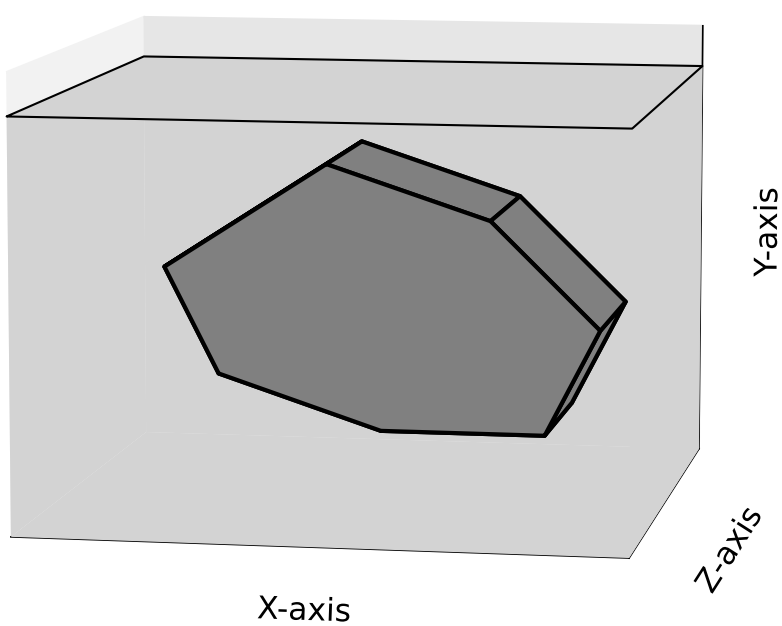

FIG. 3. Prism region and water

If the system is in hydrostatic equilibrium, the net force acting on the water inside the prism must be zero. The gravitational force acting on the water inside the prism is

$$F_w = -\rho g z_0 A. \tag{9}$$

Since the water pressure varies with depth, the average pressure on the face corresponding to the edge connecting the position vectors $\mathbf{r}_i$ and $\mathbf{r}_{i+1}$ is determined by the mean depth along that edge. This average pressure is

$$p_{i,\mathrm{av}} = \rho g \left(-\frac{y_i + y_{i+1}}{2} \right). \tag{10}$$

Now see Fig. 4. The force exerted on this face has magnitude

$$F_i = p_{i,\mathrm{av}} z_0 \|\mathbf{r}_{i+1} - \mathbf{r}_i\| = \rho g z_0 (-\frac{y_i + y_{i+1}}{2}) \|\mathbf{r}_{i+1} - \mathbf{r}_i\|. \tag{11}$$

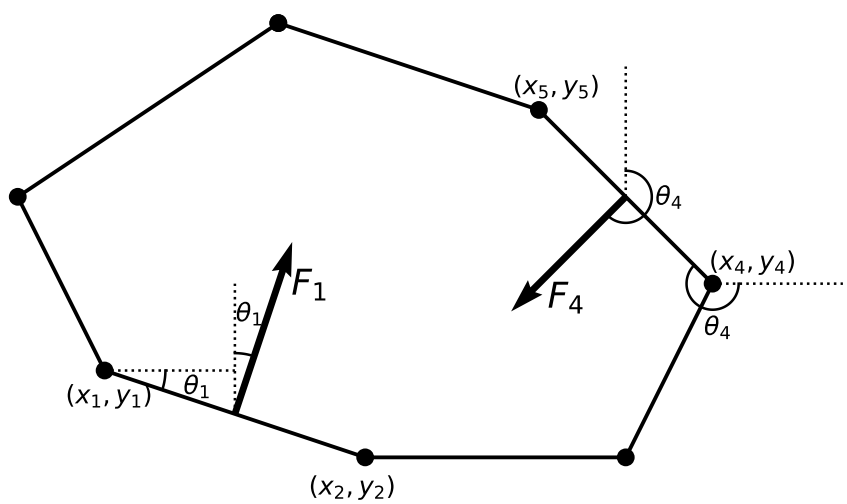

FIG. 4. Force diagram for the hydrostatic pressure

The net vertical force resulting from the hydrostatic pressure, which corresponds to the buoyant force, is

$$\begin{aligned} F_b &= \sum_{i=0}^{n-1} F_i \cos\theta_i \\ &= \sum_{i=0}^{n-1} \rho g z_0 (-\frac{y_i + y_{i+1}}{2})(x_{i+1} - x_i) \\ &= \frac{\rho g z_0}{2} \sum_{i=0}^{n-1} (x_i y_{i+1} - x_{i+1} y_i + x_i y_i - x_{i+1} y_{i+1}) \\ &= \frac{\rho g z_0}{2} \sum_{i=0}^{n-1} (x_i y_{i+1} - x_{i+1} y_i). \end{aligned} \tag{12}$$

The intermediate terms $x_i y_i - x_{i+1} y_{i+1}$ cancel out upon summation. Since the system is in equilibrium,

$$F_w + F_b = 0. \tag{13}$$

This condition is, in fact, a direct expression of Archimedes' principle (see Ref.[2], Sec. 2.7 for further discussion). Combining Eqs. (9), (12), and (13), we obtain

$$A = \frac{1}{2} \sum_{i=0}^{n-1} (x_i y_{i+1} - x_{i+1} y_i). \tag{14}$$

This completes the proof.

IV. EXTENSIONS TO CONTINUOUS SYSTEMS

A. Dido's problem

The classical optimization problem known as Dido's problem asks which closed curve of fixed length L encloses the maximum area. The isoperimetric polygon theorem provides a simple explanation: a closed curve can be viewed as a polygon with an infinite number of infinitesimal edges. For the enclosed area to be maximized, all vertices must lie on a common circle. Thus, the optimal shape must be a circle.

B. Ideal gas law

Now let Ω represent a general three-dimensional submerged volume (not necessarily a prism). The gravitational force acting on the water within Ω is

$$\mathbf{F}_w = -\rho g \, \mathrm{Vol}(\Omega) \, \hat{\mathbf{y}}, \tag{15}$$

and the buoyant force is given by

$$\mathbf{F}_b = \iint_{\partial\Omega} \rho g (-y)(-d\mathbf{A}) = \rho g \iint_{\partial\Omega} y \, d\mathbf{A}. \tag{16}$$

The condition that the net force on Ω vanishes in the hydrostatic state (i.e., Archimedes' principle) leads to

$$\rho g \iint_{\partial\Omega} y \, d\mathbf{A} = \rho g \, \mathrm{Vol}(\Omega) \, \hat{\mathbf{y}}. \tag{17}$$

This expression is often derived to justify Archimedes' principle; see, for example, Ref.[9]. Canceling the common factor ρg and taking the dot product with $\hat{\mathbf{y}}$ yields

$$\iint_{\partial\Omega} \mathbf{y} \cdot d\mathbf{A} = \mathrm{Vol}(\Omega). \tag{18}$$

By symmetry, it follows that

$$\iint_{\partial\Omega} \mathbf{x} \cdot d\mathbf{A} = \iint_{\partial\Omega} \mathbf{y} \cdot d\mathbf{A} = \iint_{\partial\Omega} \mathbf{z} \cdot d\mathbf{A} = \mathrm{Vol}(\Omega). \tag{19}$$

Summing these equations gives

$$\iint_{\partial\Omega} \mathbf{r} \cdot d\mathbf{A} = 3\,\mathrm{Vol}(\Omega), \tag{20}$$

which is a special case of the divergence theorem.

Eq. (20) is commonly used in the derivation of the ideal gas law $PV = NkT$ for a gas confined within a container of arbitrary shape (see Example 7.14 in Ref.[10]). Thus, it provides an alternative route to the ideal gas law that avoids explicit use of the divergence theorem, making the argument accessible to students unfamiliar with vector calculus.

V. DISCUSSION

The approaches presented in this paper are not mathematically rigorous proofs in the traditional sense, but rather physically-motivated derivations. Interestingly, before considering the hydrostatic argument, the author was not even aware that the maximal-area polygon theorem existed. This demonstrates how physical reasoning can provide an intuitive bridge to mathematical concepts, making abstract results more accessible to students.

VI. AUTHOR DECLARATIONS

The authors have no conflicts to disclose.

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